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Sponsored by Dr. Dylan Drotman from Neubo Inc.

Overview

We aim to introduce children aged 7-12 to soft robotics and circuitry. Using logic gates and soft actuators, our pneumatic building blocks provide a safe, hands-on learning experience for concepts related to computation, electronics, and soft robotics. These blocks allow kids to construct and control soft robots, fostering early engineering skills without electrical hazards. Neublox aims to inspire creativity and problem-solving, shaping future engineers and innovators through playful learning.



Fig 1: Air Walking Robot from Dr. Drotman's research at UCSD

Push to Break Button [1]

- Functionality:** Controls airflow by pressing the button, closing flow when button is pressed.

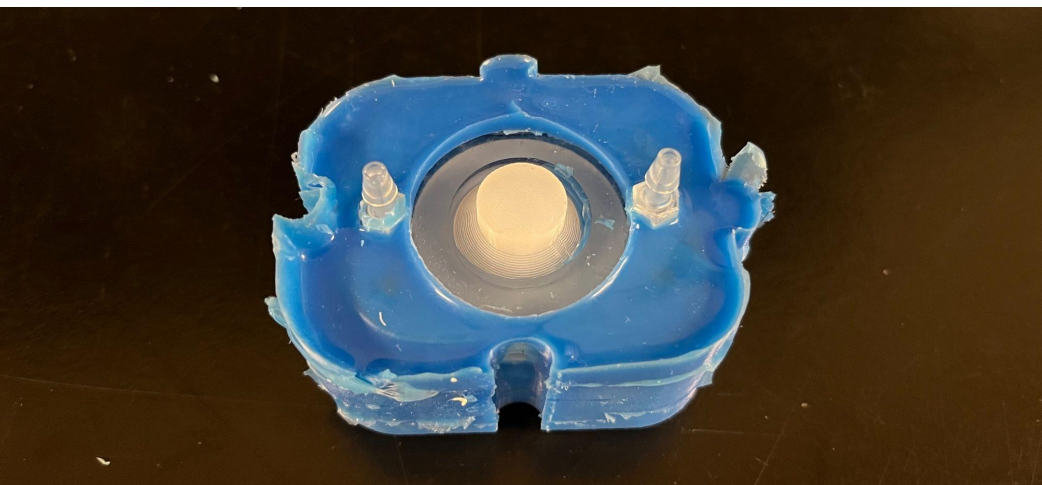


Fig 2: Push to Break Button Block

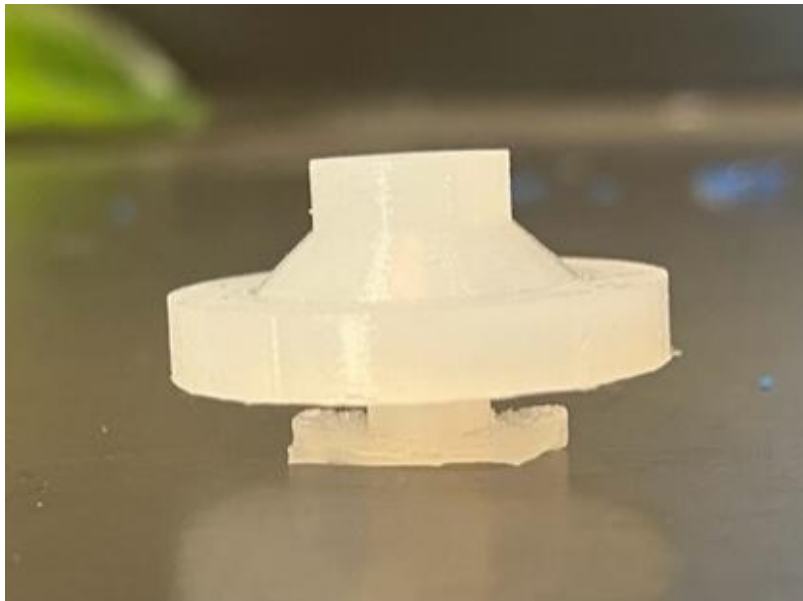


Fig 3: Silicone Button

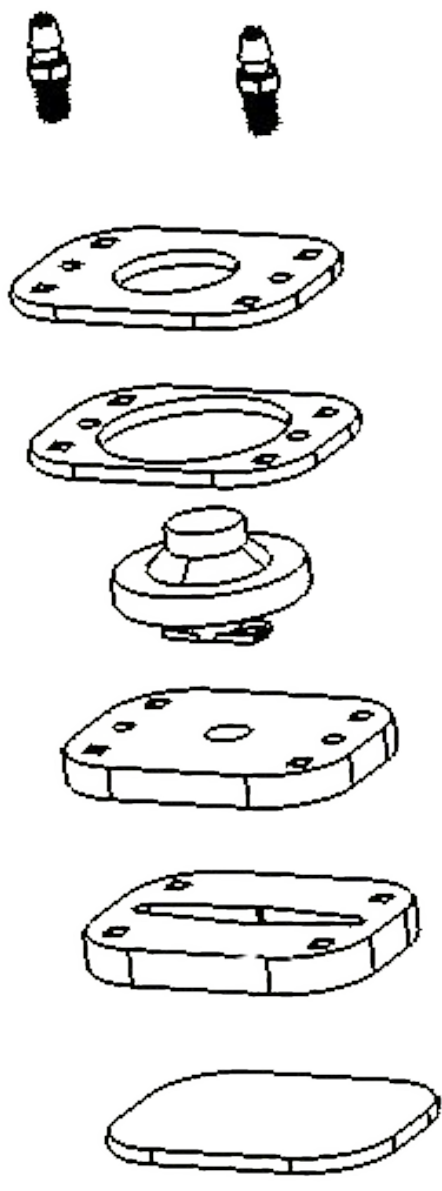


Fig 4: Button Assembly

Resistor Block

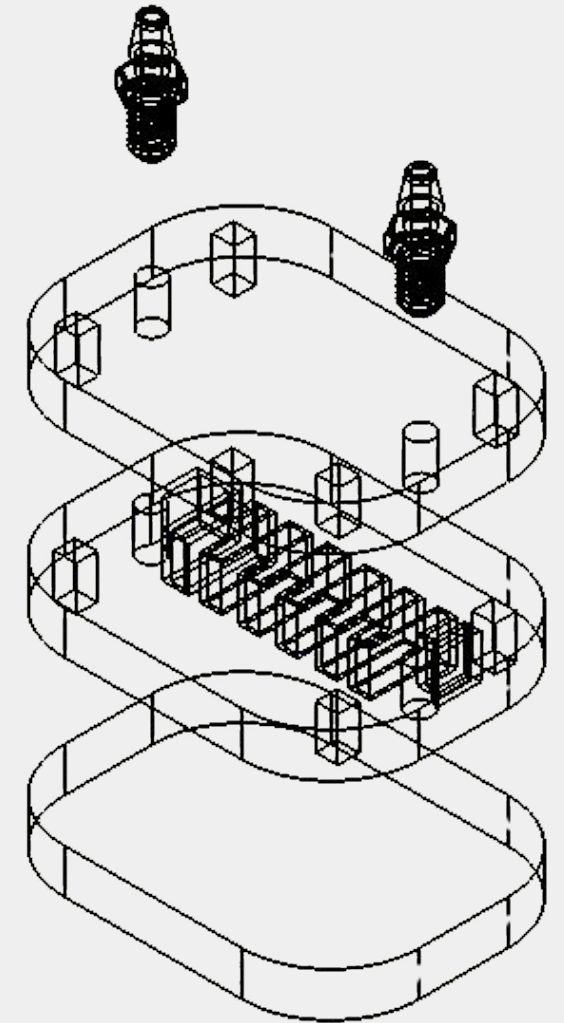


Fig 5: Resistor Assembly

- Functionality:** Control air flow to reduce systems' speed, pressure, and force.

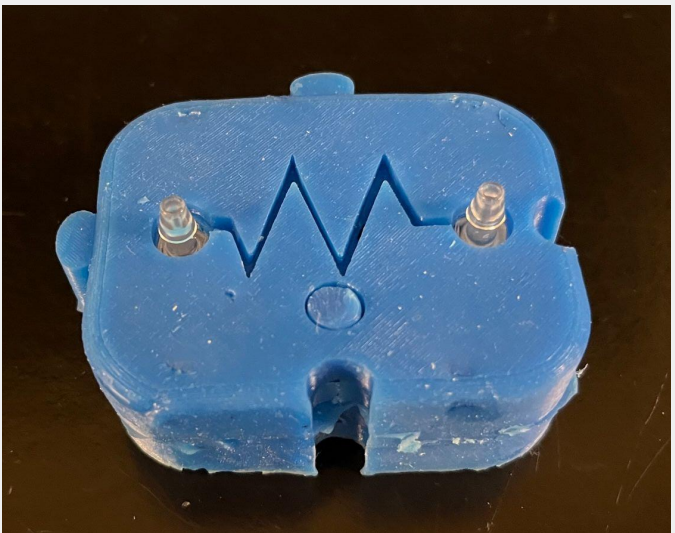


Fig 6: Resistor Block

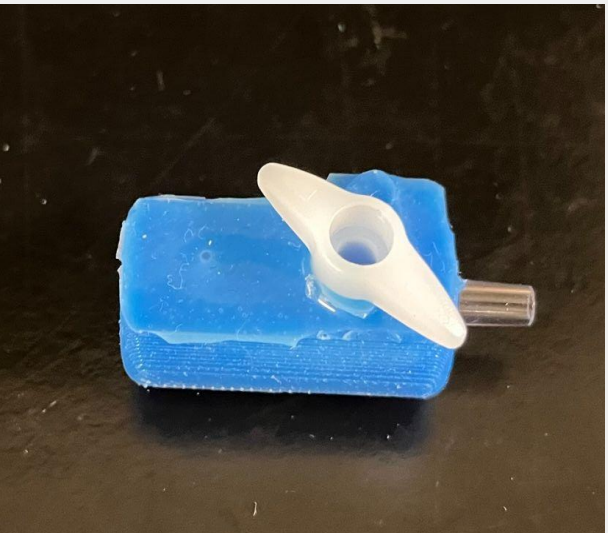


Fig 7: Variable Resistor Block

NOT Logic Gate Block

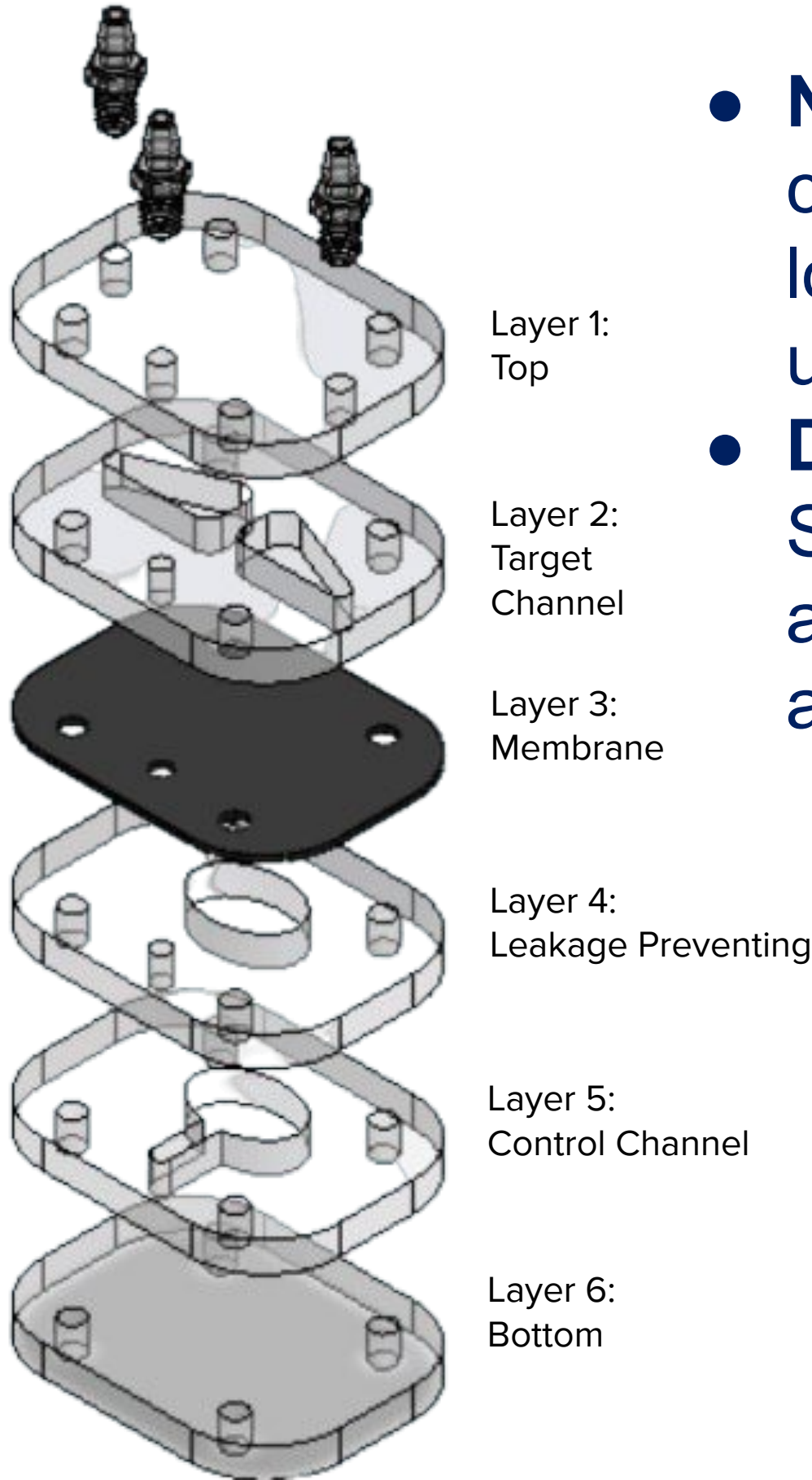


Fig 8: Pneumatic NOT Gate Exploded View

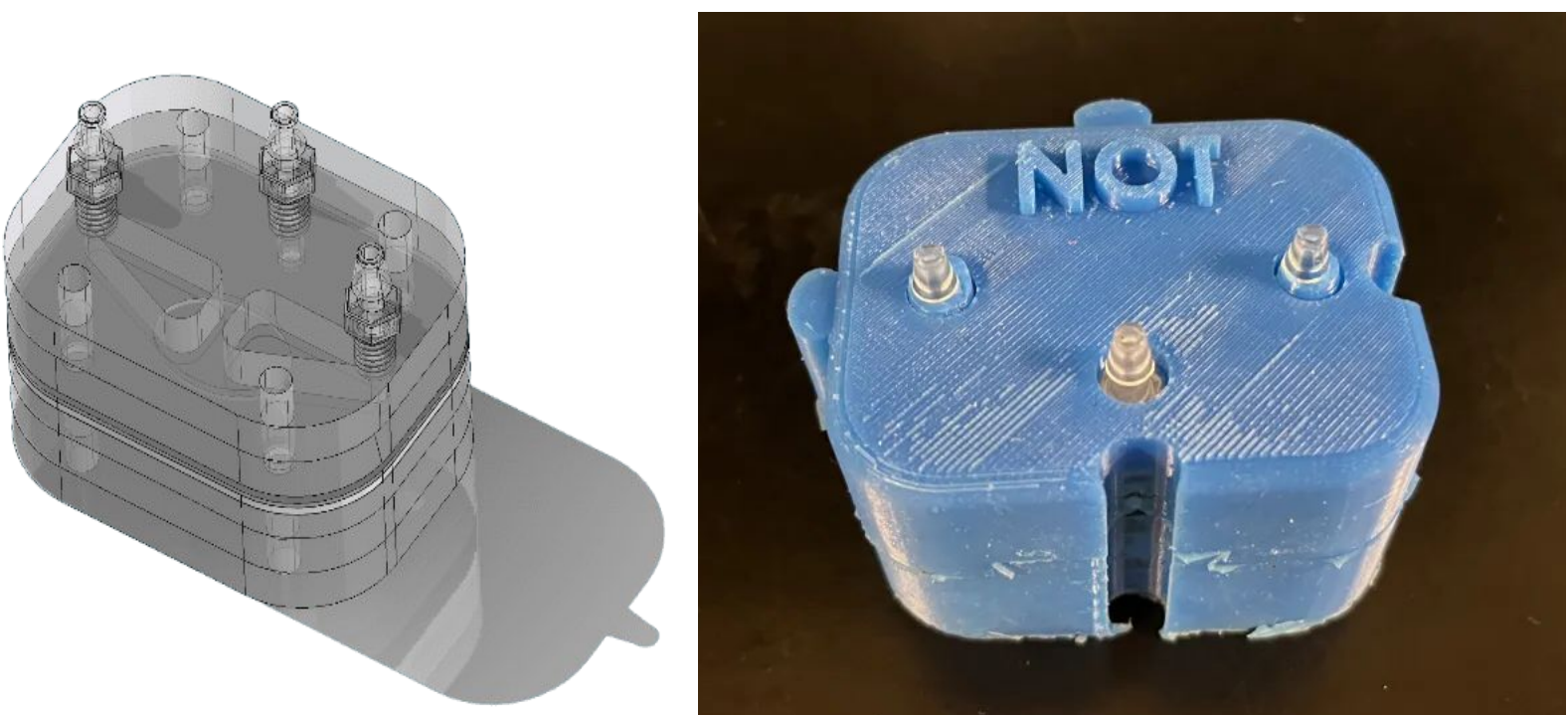


Fig 9: NOT Gate Block

- NOT Gate:** Negates flow from the control channel and enables creation of various logic gates (NAND, NOR, AND, XOR) with unique functionalities.
- Design and Assembly:** Designed in SolidWorks; assembled using acrylic layers and polyurethane membrane with adhesive.

Connector Pieces

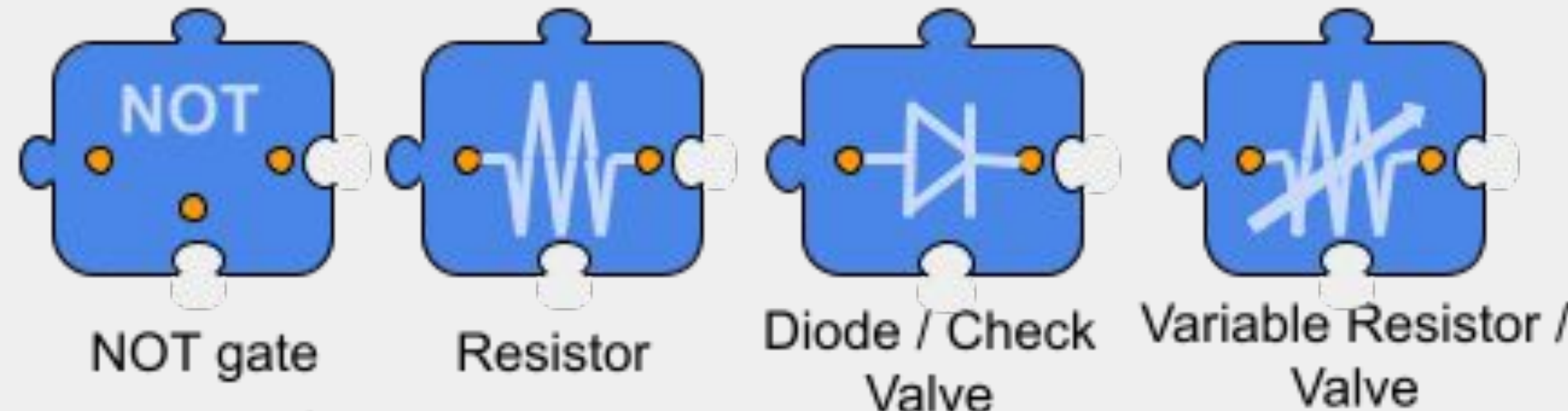


Fig 10: Multi-Directional Connector Block and Tubing

- Functionality:** Connects functional block that holds up to 10-15 Psi.
- Final Design:**
 - Silicone molded using outer diameter of the barb adapter that ensures a tight seal and less force than tubing.
 - Incorporates junction splitting functions.

Deliverables Kit

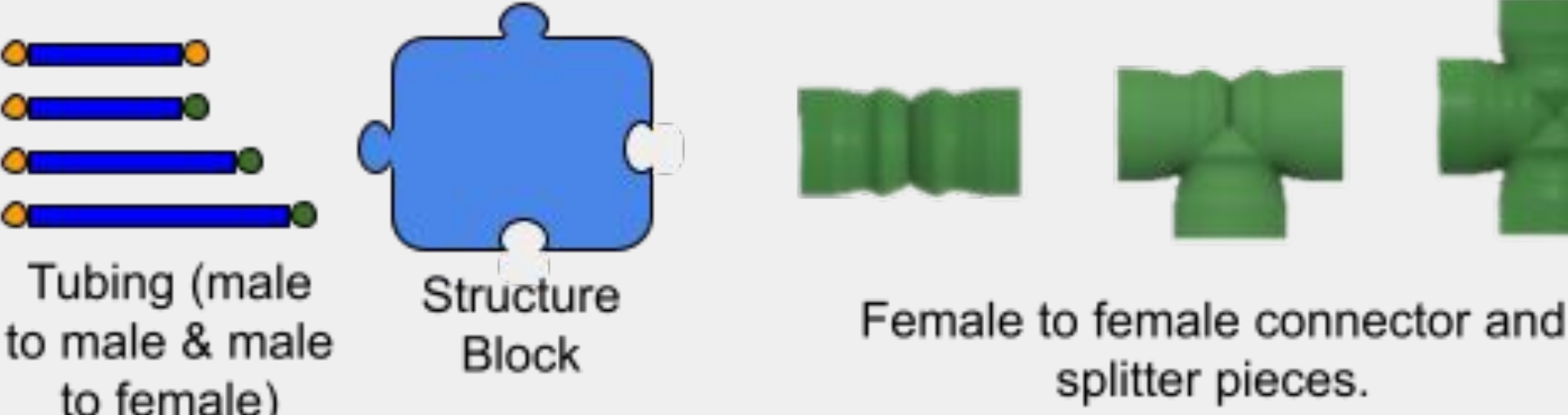
Circuit Components:



Interactive Components:



Connection Components:



Educational Circuits

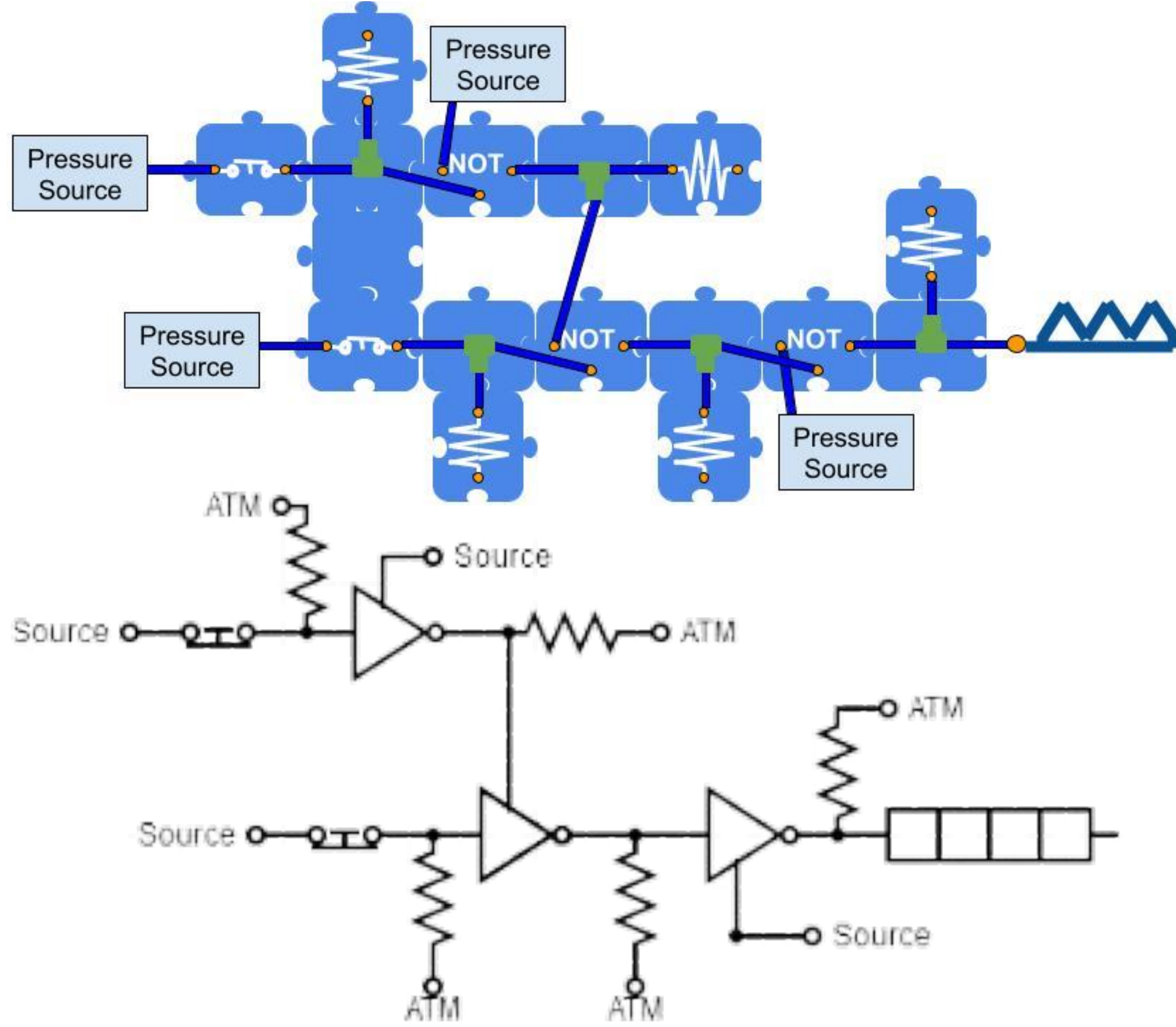


Fig 11: OR gate demonstration circuit diagram in a traditional format to the left and as a block diagram using our kit's components

Future Improvements

- NOT Gate Functionality:** Add a magnets above the target channel divide shown in layer 2 and below the layer 3 membrane to assist the controlling input in closing the target channel.
- Educational Content:** Compare our pneumatic circuit performance in controlling the arm with that of a purpose built electronic controller for pneumatics.
- Pressure Source:** Ensure that our circuits still work with a more accessible pressure source such as a diaphragm pump + soda bottle + pressure regulator.
- AND/OR Block :** Incorporate NOTs combination to a single block which allows for more complicated circuits.

Hardware Performance

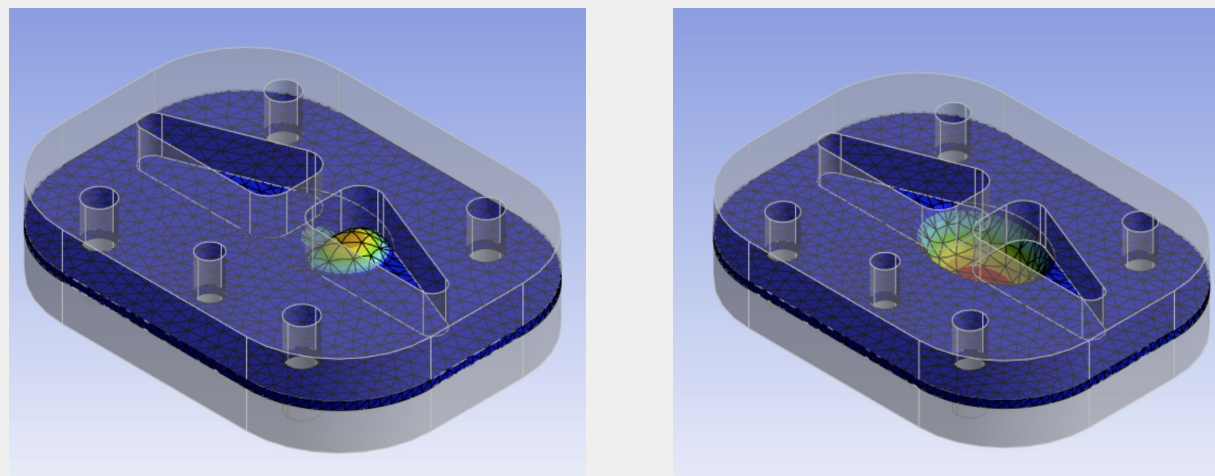


Fig 13: Membrane deflection figures for a NOT Gate from Ansys. Left is a closed gate with Control input, hence no output (1). Right is an open gate with no control input, hence there is output (2)

	(1)	(2)
Control Input (C)	1	0
Output (O)	0	1

Table. 1: NOT Gate Truth Table

Impact on Society and Safety

- Future Influence:** Neublox can inspire young engineers, fostering interest in soft robotics and electronics. This could lead to advancements in public health, infrastructure maintenance, and broader soft robotics applications.
- Safety:** To ensure the safety of the users of our product, 7-12 year olds, we have ensured that each block is soft and that our operating pressures are safe.

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- Professor Abhishek Saha & Professor Nick Gravish for your expertise and mentorship.
- Dr. Dylan Drotman for the opportunity to learn and work on an exciting project.
- Hannes Du and UCSD Makerspace for sponsor parts and materials.

References: [1] Novelchip, and Instructables. "Vacuum Powered Fluidic Ink." Instructables, Instructables, 12 Oct. 2017, www.instructables.com/Air-Powered-Fluidic-Ink-LEDs-and-Circuits/.
[2] "DIY a Food-Grade Soft Gripper for Your Delta X s Robot for Just... \$10." *YouTube*, YouTube, 13 May 2023, www.youtube.com/watch?v=Eo3y-UqJ100.

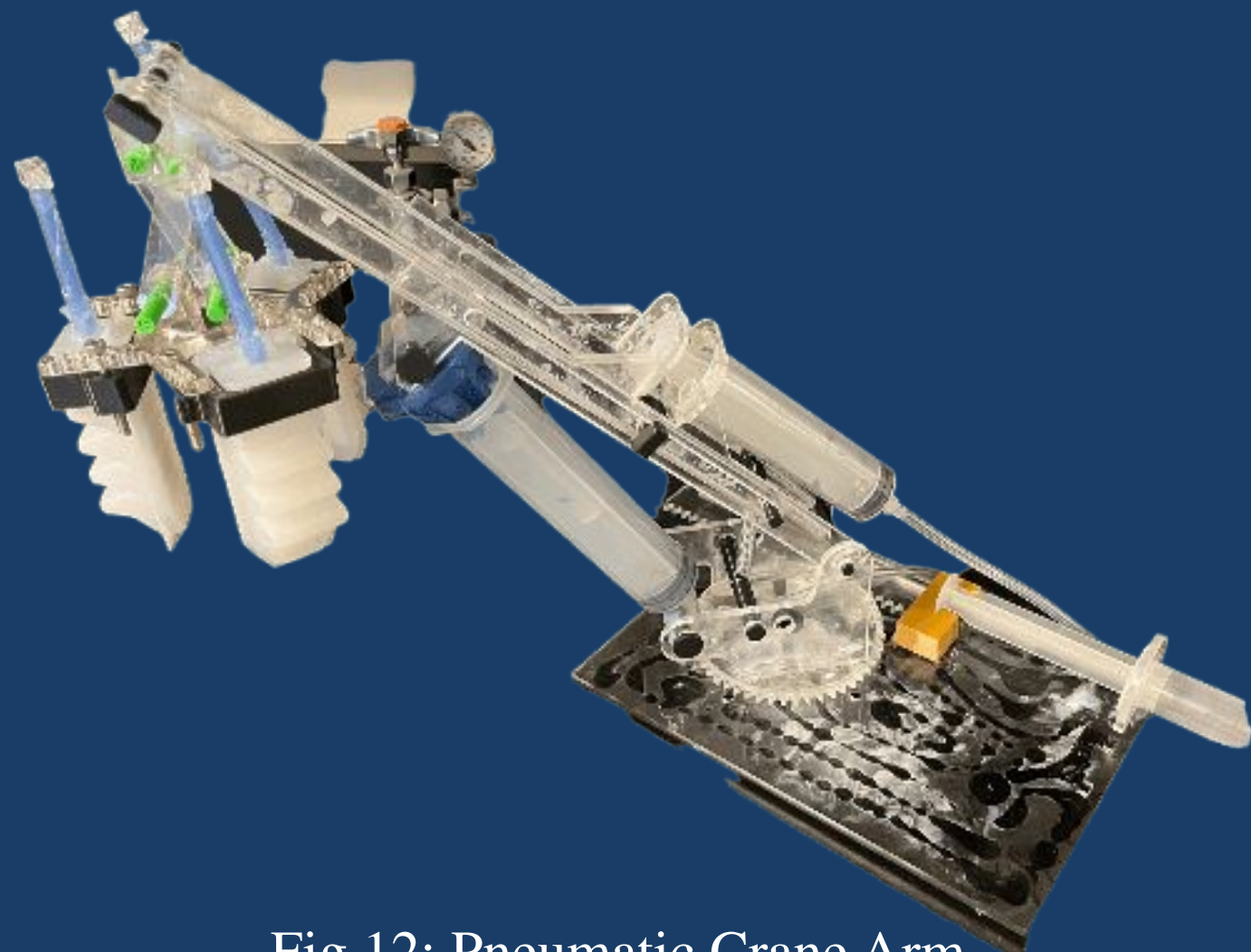


Fig 12: Pneumatic Crane Arm